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Mirror-image tests reveal hemispheric asymmetries that alter whole-brain connectome simulations

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Whole-brain connectomes allow spiking models of an entire insect brain to be built from wiring alone[1–3], and their predictions can be confirmed experimentally[3]. Yet connectomes contain reconstruction errors — missed and spurious synapses, truncated neurites and mispredicted neurotransmitters[1,4,5] — with unknown effects on simulated activity. Here we use the two hemispheres of FlyWire as built-in replicates. Simulated from the right instead of the left hemisphere, all six published benchmark predictions of a whole-brain *Drosophila* model[3] failed, and sugar-receptor stimulation drove about 8,600 of 138,639 neurons into runaway excitation. Four left–right asymmetries, probably reconstruction errors, account for most failures: weaker sugar-receptor outputs, a grooming interneuron with a quarter of its homologue's inputs, a single sugar-receptor neuron wired into the antennal lobe, and 40 antennal-lobe neurons predicted with low confidence to be serotonergic, which act as excitatory amplifiers. Correcting three of them restores four predictions. Error injection shows that predictions tolerate randomly placed spurious synapses but that most halve when 14–24% of synapses are missing. Across 131 sensory cell types, left and right responses agree far less than repeated runs (median r = 0.58 versus 0.99). Mirror tests offer a simple control for connectome-based simulations.

Main

Dense electron-microscopy reconstructions now cover the central brain[6] and the entire brain of the adult fruit fly[1,2] and, more recently, its brain and ventral nerve cord together[7,8]. These wiring diagrams have been turned directly into simulations. A leaky integrate-and-fire (LIF) model of the FlyWire brain with a single free parameter predicted which neurons respond to taste and mechanosensory input, and 91% of 164 tested predictions were confirmed experimentally[3]. Connectome-constrained models of the visual system make similar progress[9,10]. The appeal is that a wiring diagram, plus generic neuronal properties, may be sufficient to predict function, although how far anatomy constrains activity remains debated[11–13].

Wiring diagrams are, however, measurements with errors. Automated synapse detection in the full adult fly brain (FAFB) volume reaches an F-score of about 0.74 (ref. 5); only 44.7% of postsynaptic sites in FlyWire are attached to proofread neurons[1]; most tracing errors are omitted fine branches[14]; neurotransmitter identity is predicted from images with 87–94% accuracy, falling to 33% for serotonin[4]; and roughly 40% of connections found in one hemisphere are absent from the other[2]. Shiu et al. tested how their predictions depend on global parameters[3], but how specific

reconstruction errors propagate into simulated activity has not been measured, and most published simulations stimulated neurons in one hemisphere[3].

The bilateral symmetry of the fly brain offers a natural control. Homologous left and right neurons share cell types and, for strong connections, more than 90% of their partners[2], and the two hemispheres of a larval connectome have been compared as paired network samples[15]. A prediction that reflects the biology of a circuit should therefore hold when the same experiment is simulated from the mirror-image side of the same brain. Here we performed this mirror test for the benchmark predictions of the FlyWire whole-brain model, identified left–right asymmetries responsible for the failures we found, and quantified how seven types of reconstruction error alter model predictions.

Results

A mirror test for a whole-brain spiking model

We re-implemented the LIF model of Shiu et al.[3] as an event-driven simulator that runs many trials in parallel on a graphics processing unit (GPU; Methods). With the published equations and constants it reproduced the original Brian 2 implementation[16,17] neuron by neuron (Pearson $r \geq 0.999$ for 409–8,856 active neurons in three conditions; at 150 Hz the two MN9 motor neurons fired at 56.7 and 85.8 Hz versus 56.6 and 85.9 Hz; Extended Data Fig. 1a–c). A 1-s trial of the 138,639-neuron FlyWire v783 brain, with 15.1 million connections and 54.5 million synapses, took about 0.2 s. For the 362 neurons that kept their identifiers between proofreading versions v630 and v783 and fired in either, responses to the original sugar neurons of Shiu et al. correlated at $r = 0.997$, although MN9 fired 12% less in v783 (Extended Data Fig. 1d).

We defined each stimulus and readout by cell type and hemisphere so that every benchmark could be run from either side (Methods). The benchmarks were six predictions of Shiu et al. that were confirmed experimentally[3,18,19]: sugar gustatory receptor neurons (GRNs) activate the proboscis motor neuron MN9; bitter GRNs and Ir94e GRNs suppress this activation; and Johnston's organ (JO) neurons activate the antennal-grooming brain neuron aBN1 (ipsilaterally) and the antennal-grooming descending neurons aDN1 and aDN2 (contralaterally). Our stimuli comprise complete cell types — for example, all 33 left sugar GRNs of types LB3b and LB3c[20] and all 229 left JO-C and JO-E neurons — and therefore differ from the neuron sets selected by Shiu et al. (21 sugar and 70 JO-CE neurons); we examine those sets below.

Right-hemisphere predictions fail

Stimulated from the left hemisphere, the model reproduced all six benchmarks (Fig. 1). Left sugar GRNs drove MN9 from a stimulation rate of 50 Hz, reaching 58 Hz at 100 Hz; adding bitter or Ir94e GRNs suppressed this response by 89% and 78%; and JO neurons drove aBN1 at 68 Hz and the contralateral aDN1 and aDN2 at 29 and 33 Hz (Fig. 1b,e,f).

The same simulations run from the right hemisphere failed in every case (Fig. 1e,f). Right sugar GRNs (24 neurons) evoked less than 1 Hz in MN9 at 100 Hz and only 17 Hz at 300 Hz (Fig. 1b).

Instead, from 100 Hz onwards the network entered a runaway state in which 8,612 neurons fired in at least one of eight trials, 21 times the 412 neurons recruited from the left (Fig. 1a,c). Runaway began abruptly, at 59, 135 and 414 ms in three trials, after a period during which the right response resembled the left (Fig. 1d). The original Brian 2 code reproduced the runaway (8,742 active non-stimulated neurons; MN9 at 0.55 Hz), ruling out an implementation error. Right JO neurons drove the ipsilateral aBN1 at only 11 Hz, and the contralateral descending neurons not at all at any rate up to 300 Hz (Extended Data Fig. 4a). The failure depends partly on which right-hemisphere neurons are stimulated: the nine right sugar neurons that Shiu et al. selected (types LB3c and LB4b) drove MN9 without runaway at 150 Hz and above, although less strongly than their left set (29 versus 66 Hz at 150 Hz; Extended Data Fig. 4b).

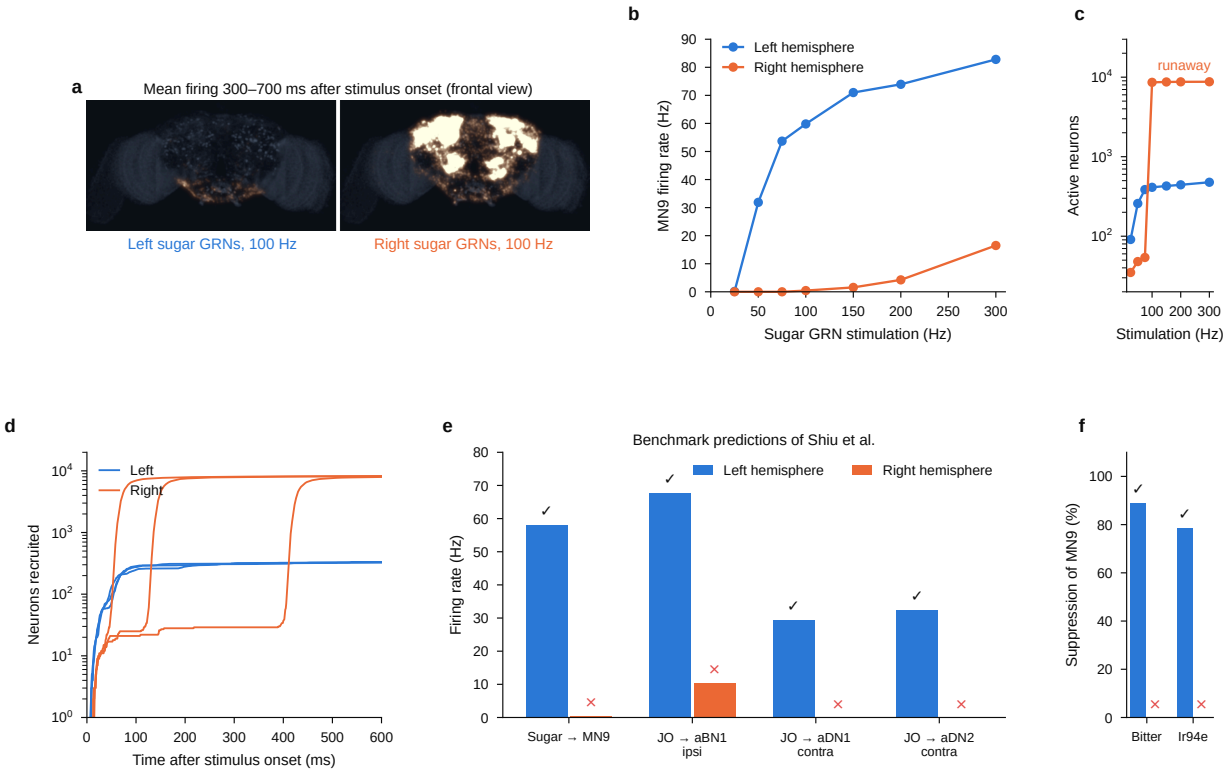


Fig. 1 | The mirror test: benchmark predictions of a whole-brain model fail when simulated from the right hemisphere. **a**, Mean firing of all FlyWire neurons (stimulated neurons excluded) 300–700 ms after the onset of 100-Hz Poisson stimulation of sugar GRNs in the left (left) or right (right) hemisphere, shown at each neuron's annotated position (frontal projection; brightness is log-scaled). **b**, MN9 firing rate (mean of the two MN9s) versus stimulation rate. **c**, Number of active neurons (firing in at least one trial) on a logarithmic axis. **d**, Cumulative number of non-stimulated neurons recruited after stimulus onset in three single trials per side. **e**,**f**, The six benchmark predictions of Shiu et al.[3] simulated from each hemisphere: activations (**e**) and suppression of sugar-evoked MN9 firing by bitter or Ir94e input, 1 − (MN9 with both)/(MN9 with sugar alone) (**f**). Ticks mark predictions reproduced by the criteria in Methods. Means of eight (**b**–**d**) or ten (**e**,**f**) trials of 1 s; gustatory stimuli at 100 Hz and JO at 150 Hz in **e**,**f**.

Under-connected sensory neurons and an under-connected readout

Right gustatory neurons differ systematically from their left homologues in the connectome. Each right sugar GRN has a mean of 125 output synapses, compared with 265 on the left (ratio 0.47); bitter, Ir94e and water GRNs show ratios of 0.64, 0.73 and 0.36, whereas JO neurons show no

deficit (1.18; Fig. 2a). Together with nine fewer annotated neurons, the right sugar GRNs provide 3,000 output synapses against 8,800 on the left.

To test whether this deficit can account for the loss of MN9 activation, we removed randomly chosen output synapses of the left sugar GRNs. MN9 activation declined steeply once the total fell below about 5,000 synapses, and at the right-hemisphere total of 3,000 synapses the left curve predicted 1–10 Hz, matching the 1.3 Hz observed from the right (Fig. 2b; both with the antennal-lobe correction described below). For this readout, the right sugar population therefore behaves like a left population with missing synapses. Because strong homologous connections in FlyWire agree closely between hemispheres[2] and improved synapse detection increases bilateral symmetry[21], a twofold deficit across a sensory population is more likely a reconstruction error, such as incompletely proofread axons or missed synapses, than a biological asymmetry, but our data cannot exclude the latter.

The grooming failure has a different source. The right aBN1, the readout of the JO benchmark, has 459 input synapses, a quarter of the 1,833 on its left homologue, and receives 35 rather than 103 synapses directly from JO neurons (Fig. 2c). Here the asymmetry lies on the readout side of the circuit.

A single wiring anomaly and mispredicted transmitters create runaway

The synapse deficit does not explain the runaway: in the published connectome, removing up to 90% of the left GRN output synapses never produced it (at most 418 active neurons). Two further asymmetries do. First, leave-one-out stimulation of the 24 right sugar GRNs identified a single neuron (LB3b, FlyWire 720575940629884119) whose omission abolished the runaway (58 instead of 8,522 active neurons; Extended Data Fig. 4c). This neuron makes 22 of its 154 output synapses onto antennal-lobe local neurons of type il3LN6, whereas left LB3b neurons make at most three such synapses (Extended Data Fig. 4d). The nine right sugar neurons selected by Shiu et al. do not include it, which explains why they did not run away.

Second, the activity entering the antennal lobe was amplified by local neurons of types lLN2T_b, lLN2T_c and lLN2T_e, after which it spread across the central brain. These neurons are predicted to release serotonin, which the model treats as excitatory[3], and their predictions are uncertain: the 40 antennal-lobe local neurons labelled serotonergic, 20 in each hemisphere, have a median confidence of 0.35, lower than that of local neurons labelled GABAergic, glutamatergic or cholinergic (Fig. 2d). Antennal-lobe local neurons are predominantly GABAergic, with a minority of glutamatergic and cholinergic cells[22], and the only serotonergic neuron known to innervate the antennal lobe is the contralaterally projecting CSD neuron[23,24], which is not a local neuron. Making only these 40 neurons inhibitory abolished the runaway (236 instead of 8,680 active neurons) without changing any left-hemisphere benchmark (Fig. 2e,f). Because this amplifier exists in both hemispheres, it is the right-specific entry route that explains why runaway occurred only from the right.

A blunter correction illustrates why sign assumptions must be treated as hypotheses. Making all 2,201 predicted serotonergic neurons inhibitory also prevented the runaway but abolished the left-

hemisphere bitter suppression (MN9 at 59 Hz without and 58 Hz with bitter input), because 7 of the 21 left bitter GRNs are themselves predicted to be serotonergic. On the right, 19 of the 21 bitter GRNs carry this label, so the same cell types receive different transmitter predictions in the two hemispheres.

Correcting three asymmetries restores four predictions

With the antennal-lobe sign correction alone, the right hemisphere still reproduced none of the six benchmarks, because the runaway was replaced by a weak response. Scaling the output weights of each right sensory population to the left per-neuron mean (×2.12 for sugar, ×1.57 for bitter and ×1.38 for Ir94e GRNs) restored MN9 activation (47 Hz) and its suppression by bitter (85%) and Ir94e (93%) input. Scaling the inputs of right readout neurons to their left totals (×3.99 for aBN1 and ×1.06 for MN9) restored activation of aBN1 by JO neurons (74 Hz versus 68 Hz on the left; Fig. 2f). Contralateral activation of aDN1 and aDN2 remained absent (3 Hz), and right sugar stimulation still recruited more neurons than left stimulation (1,044 versus 429), indicating further differences that we did not identify. These corrections test sufficiency only, using the left hemisphere as the reference.

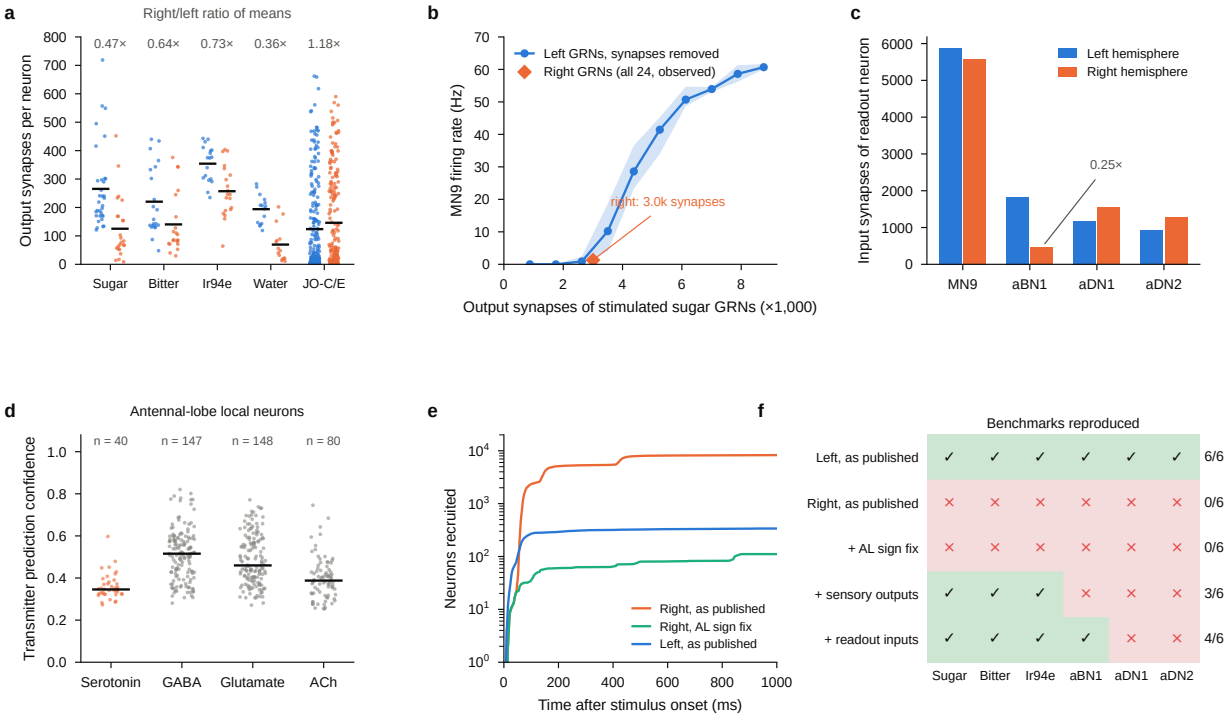


Fig. 2 | Left–right asymmetries behind the hemispheric failures. **a**, Output synapses per neuron for the stimulated sensory populations in each hemisphere; bars are means, numbers give right/left ratios. **b**, MN9 firing when random output synapses of the 33 left sugar GRNs are removed, plotted against their remaining total (blue; line, mean; band, range of three random draws), and the observed response of the 24 right sugar GRNs (diamond); 100-Hz stimulation, with the antennal-lobe (AL) sign correction applied in both. **c**, Total input synapses of the four readout neurons. **d**, Neurotransmitter prediction confidence[4] of AL local neurons by predicted transmitter; bars are medians. **e**, Non-stimulated neurons recruited after right-hemisphere sugar stimulation in the published connectome and after making the 40 serotonin-labelled AL local neurons inhibitory, with the left hemisphere for comparison (single trials). **f**, Benchmarks reproduced after cumulative corrections of the right hemisphere: AL sign correction; scaling of sensory output weights to the left per-neuron mean; scaling of readout input weights to the left totals.

Missing synapses and sign errors, not spurious synapses, degrade predictions

To measure how each kind of reconstruction error propagates into predictions, we injected seven types of error into the connectome and re-ran the left-hemisphere benchmarks (three random draws per level; Fig. 3). Predictions were insensitive to randomly placed spurious synapses: adding 80% more synapses between partners drawn in proportion to their existing connectivity left every benchmark unchanged (Fig. 3d). They were sensitive to missing synapses. Removing synapses at random halved the activation of aDN1 at 14%, of MN9 at 21% and of aBN1 at 45%; at 30% the MN9 response fell to 8% of intact, and at 40% it was abolished (Fig. 3a,h). Truncating dendrites or axons — removing a random fraction of each neuron's inputs or outputs — had similar effects (Fig. 3b,c). Bitter and Ir94e suppression remained intact for as long as sugar still drove MN9, beyond which they could not be evaluated.

Random sign errors were tolerated at low rates. When 1–2% of neurons switched between excitatory and inhibitory, mean benchmark responses stayed within 15% of intact, although single draws deviated by up to 50%; at 8–18%, four of the five benchmarks lost half of their effect, and at 20% two of three draws ran away (Fig. 3e). For errors drawn from the transmitter classifier's own confidence, a scale that changed the transmitter of 8.2% and the sign of 6.1% of neurons — comparable to the classifier's reported per-neuron error of about 6% (ref. 4) — left all benchmarks within 20% of intact in two of three draws, whereas the third ran away and lost most aDN1 activation (Fig. 3f). Such calibrated errors were not more damaging on average than random sign errors of similar size, but they triggered runaway more often (one of three draws versus none at 5% random flips). Variability of synaptic strength around the synapse count, with the mean weight held constant, was largely tolerated at a log-normal σ of 0.25 (aDN1 activation −25%); at 0.5, aDN1 activation fell by 56% and one of three draws ran away, and at $\sigma = 1$ every draw ran away (Fig. 3g). Analysis conventions mattered as well: discarding connections with fewer than five synapses, a common threshold[1], reduced JO-evoked activation of aDN1 by 83% and of aBN1 by 46%, and treating glutamate as excitatory drove 30,128 neurons into runaway and reversed both suppression benchmarks, as anticipated by Shiu et al.[3] (Extended Data Fig. 2b,c). Brain-wide similarity to the intact response declined gradually even where benchmarks failed abruptly (Extended Data Fig. 2a).

These dose–response curves help explain why the asymmetries of Fig. 2 matter. The model's single weight parameter was tuned to the connectome as reconstructed[3], so a deficit shared uniformly by the whole brain would be partly absorbed by that calibration, whereas a deficit confined to one circuit is not. Removing synapses only from the left sugar-receptor neurons halved the MN9 response at a loss of about 50% (Fig. 2b), and the right sugar-receptor neurons provide 66% fewer output synapses in total than the left.

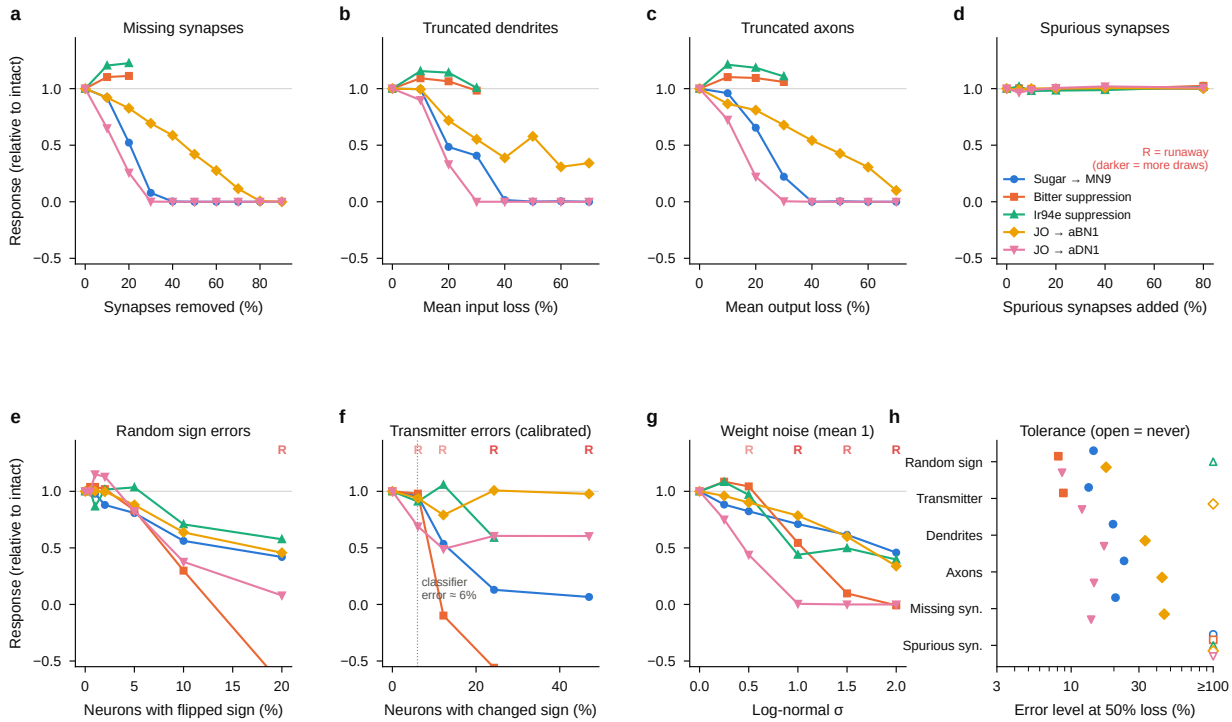


Fig. 3 | Dose–response of benchmark predictions to seven types of reconstruction error. Left-hemisphere benchmarks after injecting errors into the FlyWire connectome, relative to the intact connectome (mean of three random draws, each of eight trials). Suppression is shown only where sugar alone still drove MN9 above 20 Hz; values below zero indicate facilitation. **a**, Synapses removed independently at random. **b**,**c**, Each neuron loses a random fraction (beta-distributed, mean shown) of its input (**b**) or output (**c**) synapses. **d**, Spurious synapses added between partners drawn in proportion to their existing connectivity. **e**, Random neurons switch between excitatory and inhibitory. **f**, Transmitter identities resampled according to the classifier's confidence[4]; the x axis shows the resulting fraction of neurons whose sign changed, and the dotted line marks the classifier's reported per-neuron error rate. **g**, Log-normal noise with mean 1 multiplied onto each connection weight. R marks levels at which at least one draw ran away (more than 2,000 active neurons in any of the four conditions; darker, more draws). **h**, Error level at which each benchmark fell to half of its intact value; open symbols, never reached within the tested range; suppression benchmarks are omitted where they could not be evaluated, and weight noise (in units of σ) is not shown.

A brain-wide mirror screen

To ask whether hemispheric disagreement is specific to the benchmark circuits, we stimulated each of 131 sensory cell types — every type with at least three annotated neurons on each side — from the left and from the right (Methods). In the published connectome, 60 types (46%) drove more than 2,000 non-stimulated neurons from at least one side, including 52 of the 53 olfactory receptor neuron types (Fig. 4a). After the 40-neuron antennal-lobe sign correction, this occurred for 8 olfactory types (6%), and after the additional cell-type consensus correction, for one type, photoreceptors R1–6, whose 4,029–4,423 stimulated neurons per side drive much of the optic lobes. The antennal-lobe amplifier therefore accounts for most of the runaway behaviour of the published model.

Even without runaway, mirror-image stimulation produced markedly different brain-wide responses. We compared, cell type by cell type, the responses evoked from the left and from the right with those obtained when the same side was stimulated twice with new random input. Excluding types that ran away, repeated stimulation gave nearly identical responses (median r = 0.99), whereas left

and right stimulation agreed far less (median r = 0.58, interquartile range 0.43–0.72; $P = 4 \times 10^{-21}$, two-sided Wilcoxon signed-rank test, n = 118 types; Fig. 4b). For 85 of these 118 types, mirror agreement was lower than the lowest agreement between repeats. Discordance was not predicted by the asymmetry of the stimulated population's own synapse totals (Spearman ρ = −0.01, P = 0.9; Fig. 4c) or of its predicted transmitter signs (ρ = −0.06, P = 0.5), and making transmitter signs consistent within every cell type improved it only slightly (median r = 0.62 versus 0.58, P = 0.01, n = 115). One systematic component was lateral: for olfactory types, the share of evoked activity in the stimulated hemisphere was lower for right than for left stimulation (median 0.45 versus 0.53, P = 0.002, n = 45), whereas mechanosensory types showed no such bias (0.89 versus 0.87, P = 0.44, n = 39; Fig. 4d).

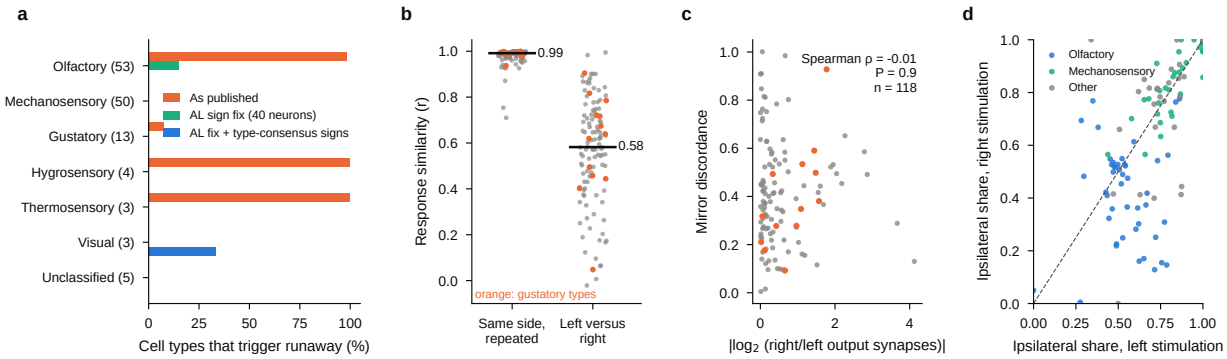


Fig. 4 | A mirror screen of 131 sensory cell types. Each type was stimulated at 100 Hz from either hemisphere (four trials). **a**, Percentage of types in each class (number of types in brackets) that drove more than 2,000 non-stimulated neurons from at least one side, in the published connectome, after the antennal-lobe (AL) sign correction and after the additional cell-type consensus correction. **b**, Similarity of mirror-aligned, cell-type-averaged responses (Pearson r of log rates) between two independent runs from the same side and between left and right stimulation, after the AL correction; bars, medians; orange, gustatory types. **c**, Mirror discordance (repeat similarity minus left–right similarity) versus the absolute $\log_2$ ratio of the stimulated population's right and left output synapse totals. **d**, Share of evoked activity in the stimulated hemisphere for left versus right stimulation; mirror symmetry predicts points on the diagonal. Types that ran away are excluded from **b**–**d**.

Discussion

Simulating the same experiment from the two hemispheres of one connectome is a control that requires no new data and no assumption beyond bilateral symmetry. Applied to a whole-brain model whose predictions had been validated experimentally[3], it showed that all six benchmarks, as defined here by complete cell types, would not have been reproduced from the right hemisphere. The failures were not subtle: the right hemisphere failed to drive a motor neuron that the left drove at nearly 60 Hz and instead recruited 6% of the brain in runaway excitation.

The asymmetries we identified suggest practical checks. Whole sensory populations can be under-connected, as were the right gustatory receptor neurons; comparing per-neuron synapse totals of stimulated populations between hemispheres is trivial and would have flagged this. Readout neurons can be under-connected, as was the right aBN1. A single neuron with unusual connections can open a path into another circuit, and neurotransmitter predictions of low confidence can turn that circuit into an amplifier; in a model without adaptation or other stabilizing mechanisms, the combination produced runaway. Network-wide activity should therefore be

monitored in every simulation, not only the readout neurons, and conclusions that depend on low-confidence transmitter predictions — especially serotonin, the transmitter predicted least accurately[4] — should be tested by varying those signs.

The dose–response analysis indicates where reconstruction effort matters most for simulation. Randomly placed spurious synapses were harmless even at high rates, presumably because they are scattered and weak, whereas missing synapses degraded predictions steeply, so completeness matters more than precision. Missing synapses are also the dominant error in automated reconstructions[1,5]. Sign errors at the rate implied by the classifier's accuracy left most predictions intact but could trigger runaway, and their effect depended on which neurons were affected: 40 low-confidence predictions in one circuit mattered more than thousands of random flips elsewhere. The brain-wide screen shows that the benchmark failures were not isolated. Responses to most sensory types differed between hemispheres far more than between repeated runs, and the discordance tracked neither the synapse totals nor the predicted signs of the stimulated neurons, which suggests that it arises from asymmetries distributed through downstream circuits. A whole-brain prediction in this model is thus partly a property of the hemisphere in which it is computed. Comparable cross-connectome replications are beginning to appear for the ventral nerve cord[25] and for other uses of this model[26], and mirror tests could accompany them at no cost.

Our analysis has limitations. We cannot know which hemisphere is closer to the biological truth; the left side may contain errors that happen to favour the published predictions, and our corrections rescale weights coarsely rather than restore missing synapses. Whether the right LB3b neuron's connections to the antennal lobe reflect a merge or synapse-assignment error or genuine anatomy is unknown, and the antennal-lobe sign correction, although supported by anatomy[22–24], should be confirmed by immunostaining or transcriptomics of the lLN2T types. Benchmark outcomes depend on how stimuli are defined and on stimulation rate (Extended Data Fig. 4a,b). The model omits gap junctions, neuromodulation, adaptation and cell-type-specific membrane properties[3,11]; these simplifications apply equally to both hemispheres but may amplify differences between them. Newer synapse predictions that improve bilateral symmetry[21] and the brain-and-nerve-cord connectomes[7,8] offer independent replicates against which the mirror test can be repeated.

Connectome-based simulations are becoming a standard way to generate hypotheses about neural circuits. Our results suggest a simple practice for them: run every prediction from both hemispheres, report the agreement, monitor brain-wide activity, and treat sign assumptions for low-confidence neurons as parameters to be varied. Disagreement between the two halves of the same brain may indicate where the wiring diagram, rather than the biology, is shaping a prediction.

Methods

Connectome data

We used the FlyWire v783 connectome[1,2] as distributed with the model of Shiu et al.[3,17]: 138,639 neurons and 15,091,983 connections carrying 54,492,922 synapses, each connection with its synapse count and the predicted sign of its presynaptic neuron. A neuron is inhibitory if its predicted transmitter[4] is GABA or glutamate and excitatory otherwise (acetylcholine, dopamine, serotonin and octopamine). Cell types, hemisphere, predicted transmitter, its confidence and positions were taken from the FlyWire annotations[2] (Supplementary file 1, flyconnectome/flywire_annotations, accessed 27 October 2026). Hemisphere labels follow these annotations; the sugar neurons that Shiu et al. called right-hemisphere are labelled left in this release. For the version comparison we used FlyWire v630 (127,400 neurons) from the same repository.

Whole-brain model

The model follows Shiu et al.[3]. Each neuron obeys $\mathrm{d}v/\mathrm{d}t = (v_0 - v + g)/\tau_\mathrm{m}$ and $\mathrm{d}g/\mathrm{d}t = -g/\tau_\mathrm{s}$, with resting and reset potential $v_0 = -52$ mV, threshold −45 mV, $\tau_\mathrm{m} = 20$ ms, $\tau_\mathrm{s} = 5$ ms and a refractory period of 2.2 ms during which v and g are not integrated. At a spike, v is reset and g set to zero. A presynaptic spike increments g by $n \times w_\mathrm{syn}$ after 1.8 ms, where n is the signed synapse count and $w_\mathrm{syn} = 0.275$ mV. Stimulated neurons receive Poisson input that adds $250 \times w_\mathrm{syn}$ to v and have no refractory period. Neurons have no spontaneous activity. Trials last 1 s with a time step of 0.1 ms.

Our simulator (Python 3.12, PyTorch 2.14 with CUDA 13.0, one NVIDIA RTX 5070 Ti) integrates the linear subthreshold dynamics exactly, applies the update order of Brian 2 (integration, threshold, synaptic events, reset) and runs a batch of independent trials in parallel; 30 trials of 1 s took 6 s. Synaptic input is delivered event by event from a compressed column representation of the weight matrix, so the cost scales with the number of spikes. Comparing voltage traces with Brian 2 showed that synaptic input arriving during the refractory period does not persist in the original implementation; we reproduced this behaviour. For validation we ran the original code[17] (Brian 2.10 with Cython code generation[16]) with identical stimuli for 10–30 trials and compared firing rates neuron by neuron, excluding stimulated neurons.

Stimuli, readouts and benchmarks

Populations were defined by FlyWire cell type and hemisphere: sugar GRNs, LB3b and LB3c (33 left, 24 right neurons); bitter GRNs, LB1a–d (21, 21); Ir94e-like GRNs, LB1e and LB2a–c (18, 20), following the type composition of the Ir94e set of Shiu et al.; water GRNs, LB3a (15, 15); and JO neurons, all JO-C and JO-E types (229, 204). Readouts were MN9 (type CB0701), aBN1 (SAD093), aDN1 (DNg62) and aDN2 (DNge078), one neuron per hemisphere each. The original neuron sets of Shiu et al. were mapped to v783 by root identifier (20 of 21 left and 9 of 10 right sugar neurons present). Gustatory benchmarks used 100-Hz stimulation and JO benchmarks 150 Hz, rates at which the left hemisphere reproduced all benchmarks; rate dependence is shown in Fig. 1b and Extended Data Fig. 4a. A neuron was active if it fired at least once in any trial of a

condition, so counts of active neurons grow slightly with the number of trials. A benchmark was counted as reproduced when MN9 fired above 20 Hz for sugar; suppression, 1 − (MN9 with sugar and bitter or Ir94e)/(MN9 with sugar alone), exceeded 50% and sugar alone was reproduced; ipsilateral aBN1 fired above 20 Hz; and contralateral aDN1 and aDN2 fired above 10 Hz. A condition was classed as runaway when more than 2,000 neurons were active. Runaway onset was read from cumulative recruitment curves as the time of the steep rise beyond the plateau reached by left-hemisphere stimulation. The route of the runaway was traced by ordering neurons by the time of their first spike in single trials.

Asymmetry analyses and corrections

Output and input synapse totals were computed from the connection table. For targeted deletion, each output synapse of the left sugar GRNs was removed independently with probability p (binomial thinning of connection counts; three random draws per p). For leave-one-out stimulation, each of the 24 right sugar GRNs was omitted in turn (four trials each). The antennal-lobe sign correction made all neurons of cell class ALLN (antennal-lobe local neurons) with predicted transmitter serotonin inhibitory (40 neurons). For input compensation, the weights of all output connections of each right stimulus population were multiplied by the ratio of left to right mean output synapses per neuron when this ratio exceeded 1; for readout compensation, the weights of all input connections of a right readout neuron were multiplied by the ratio of left to right input totals when it exceeded 1 (aBN1, ×3.99; MN9, ×1.06). The cell-type consensus correction assigned to every typed neuron the sign of the confidence-weighted majority of its type's transmitter predictions, pooled over both hemispheres.

Error models

Errors were applied to the intact v783 connection table and the left-hemisphere benchmarks were re-run (sugar, sugar with bitter, sugar with Ir94e at 100 Hz; JO at 150 Hz; eight trials each), with three random draws per error level. *Missing synapses*: each synapse was removed independently with probability p. *Truncated dendrites* or *axons*: each neuron lost a fraction f of its input (or output) synapses, with f drawn from a beta distribution with mean p and concentration 4, so that losses were correlated within neurons. *Spurious synapses*: q times the total number of synapses was added as single synapses between partners drawn in proportion to their existing output and input synapse totals; added synapses took the sign of their presynaptic neuron. *Random sign errors*: a fraction r of neurons with outputs switched between excitatory and inhibitory. *Calibrated transmitter errors*: each neuron kept its predicted transmitter with probability $1 - s(1 - c)$, where c is the prediction confidence[2,4] and s a scale factor, and otherwise received a different transmitter drawn in proportion to transmitter frequencies; scales of 0.25, 0.5, 1 and 2 changed the sign of 6.1%, 12.2%, 24.3% and 47.0% of neurons (fractions of all neurons, averaged over draws), and at 0.25 the transmitter of 8.2%. *Weight noise*: each connection weight was multiplied by log-normal noise with mean 1 and standard deviation σ of the underlying normal distribution. We also tested discarding connections with fewer than k synapses and treating glutamatergic neurons as excitatory. Responses are expressed relative to the mean of three intact runs. Suppression was evaluated only when sugar alone drove MN9 above 20 Hz. The tolerance of a benchmark is the

error level, interpolated linearly, at which its mean fell below half of its intact value. Brain-wide similarity is the Pearson correlation between log(1 + rate) of the perturbed and intact responses over neurons active in either, averaged over the four conditions.

Sensory mirror screen

We selected the 131 sensory cell types with at least three annotated neurons in each hemisphere (53 olfactory, 50 mechanosensory, 13 gustatory, 4 hygrosensory, 3 thermosensory, 3 visual and 5 unclassified types). Each type was stimulated at 100 Hz from each hemisphere separately (four trials) in the published connectome, after the antennal-lobe sign correction and after the additional cell-type consensus correction. Runaway was assessed on non-stimulated neurons. For each stimulation, responses were averaged by cell type and by side relative to the stimulus (ipsilateral or contralateral), giving a mirror-aligned response vector. Mirror concordance is the Pearson correlation between log(1 + rate) of the left- and right-stimulation vectors over entries active in either; types with fewer than three active entries were excluded. To estimate the agreement expected from trial-to-trial variability alone, each stimulation was repeated with new Poisson input, and within-side reproducibility was computed in the same way and averaged over the two sides. Mirror discordance is within-side reproducibility minus mirror concordance. The ipsilateral share is the summed firing of non-stimulated neurons in the stimulated hemisphere divided by that in both hemispheres.

Statistics and reproducibility

Each condition was simulated in 4–30 independent trials with different Poisson input; error models were drawn with three random seeds per level. Firing rates are trial means. Correlations are Pearson coefficients unless stated; Wilcoxon signed-rank tests are two-sided. No statistical method was used to predetermine sample size. All analyses are deterministic given the random seeds recorded in the code.

Video

Supplementary Video 1 shows whole-brain activity from recorded spike trains together with a kinematic rendering of the NeuroMechFly v2 body[27] (flygym 2.1.1, MuJoCo 3.16) whose proboscis extension follows the simulated MN9 rate.

Data and code availability

FlyWire v783 connectivity and annotations are publicly available[1,2,17]. All code, intermediate results and figure scripts are in the project folder `fly_connectome_robustness` and will be deposited in a public archive with a DOI upon publication.

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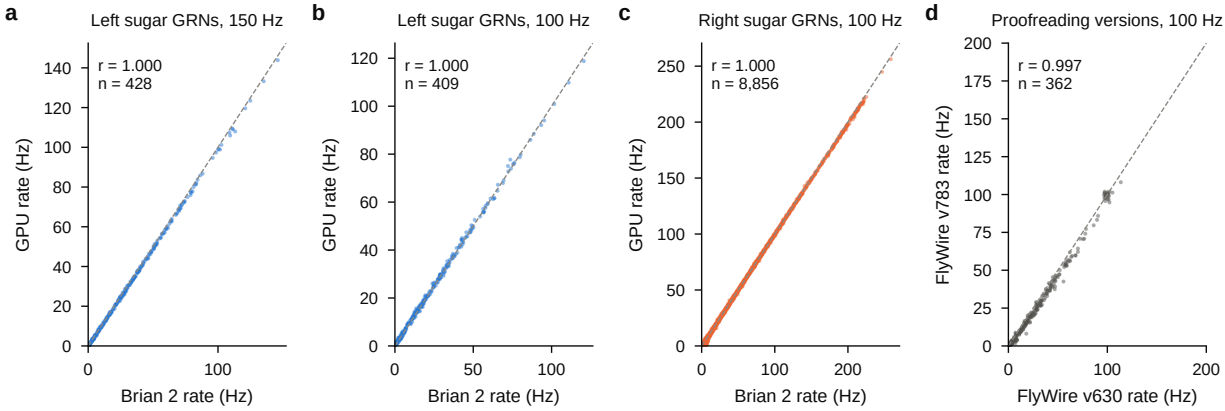
Author contributions

K.T. conceived the study, performed the analyses and wrote the paper.

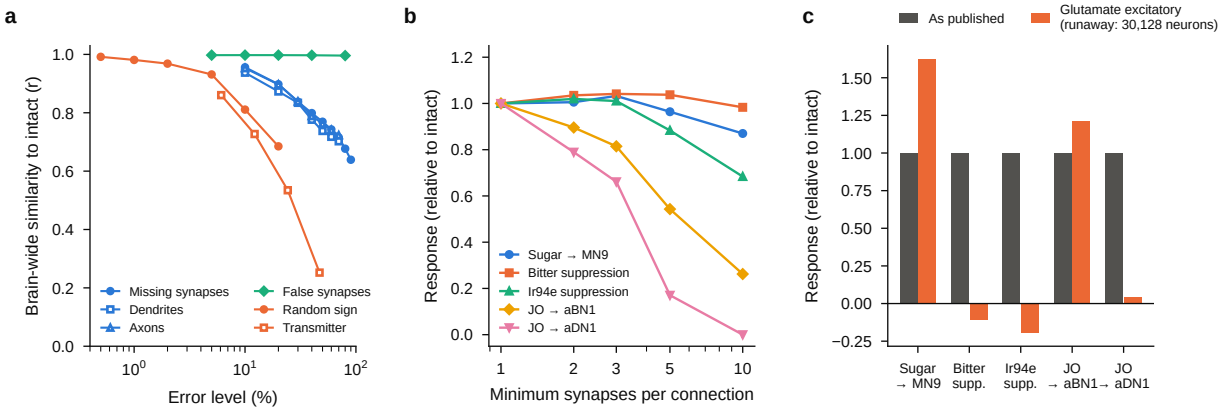
Competing interests

The author declares no competing interests.

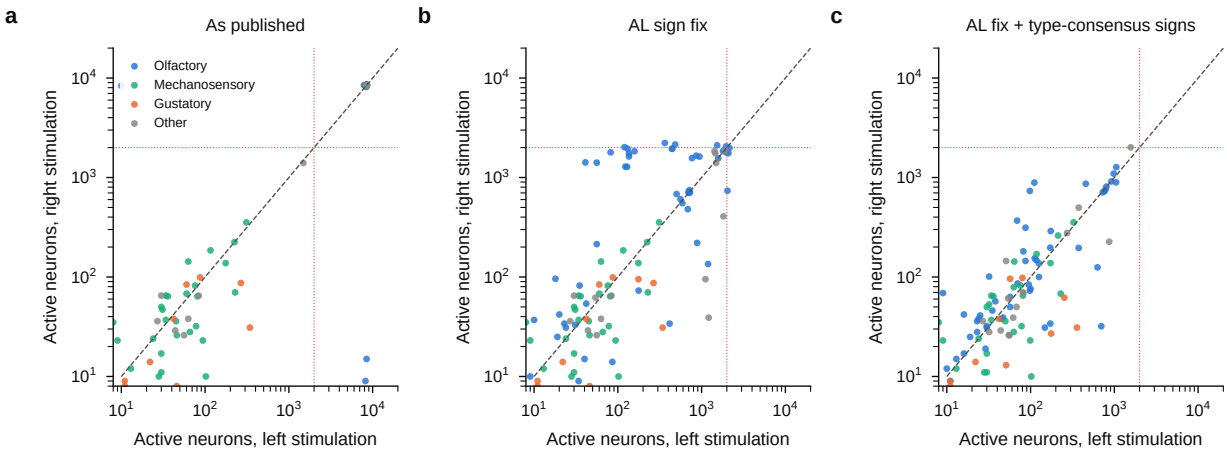
Extended Data



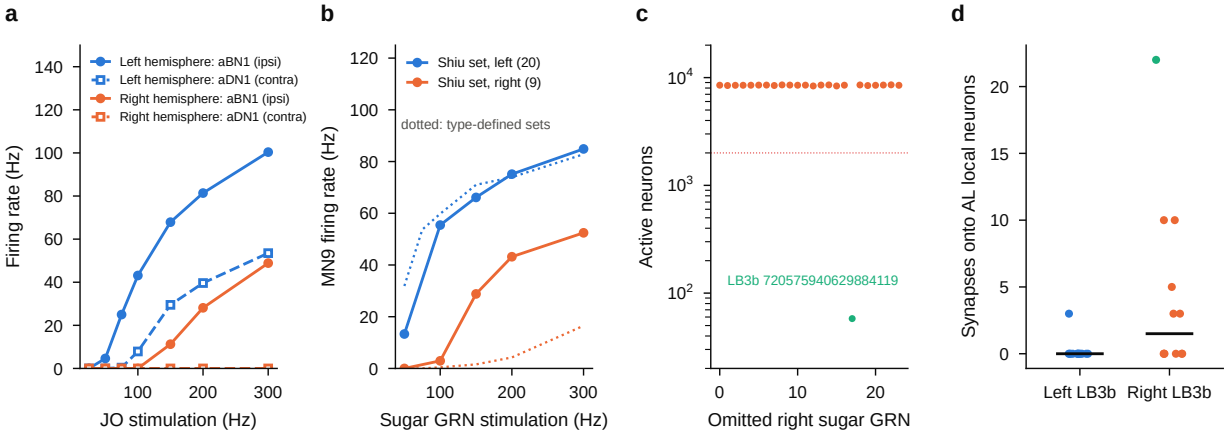
Extended Data Fig. 1 | Validation of the GPU simulator and stability across proofreading versions. a–c, Firing rate of every active non-stimulated neuron in the GPU simulator versus the original Brian 2 implementation[3,17] for identical stimuli (FlyWire v783). **a**, Left sugar GRNs at 150 Hz (30 trials each). **b**, Left sugar GRNs at 100 Hz (10 trials in Brian 2, 30 on the GPU). **c**, Right sugar GRNs at 100 Hz, a runaway condition (10 and 30 trials). **d**, Responses to the original sugar neurons of Shiu et al. (21 in v630; 20 present in v783) in the two versions, for the 362 neurons that kept their identifiers and fired in either, including 20 stimulated neurons. Dashed lines, identity.



Extended Data Fig. 2 | Brain-wide similarity, synapse thresholds and the glutamate sign assumption. a, Similarity between perturbed and intact brain-wide responses (Pearson *r* of log rates over neurons active in either) for each error type; transmitter errors are plotted at the resulting fraction of neurons with changed sign. **b**, Benchmarks after discarding connections with fewer than *k* synapses. **c**, Benchmarks when glutamatergic neurons are treated as excitatory, relative to the published sign assignment; suppression benchmarks reverse and 30,128 neurons fire in response to sugar.



Extended Data Fig. 3 | Active neurons in the sensory mirror screen. Number of active non-stimulated neurons after 100-Hz stimulation of each of 131 sensory cell types from the left (x) and the right (y) hemisphere in the published connectome (**a**), after the antennal-lobe sign correction (**b**) and after the additional cell-type consensus correction (**c**). Dotted lines, runaway criterion (2,000 neurons); dashed line, identity.



Extended Data Fig. 4 | Stimulation rate, neuron selection and the right-hemisphere entry route. a, Activation of ipsilateral aBN1 (filled) and contralateral aDN1 (open) by JO neurons of each hemisphere versus stimulation rate. **b**, MN9 firing evoked by the sugar-neuron sets of Shiu et al. (left, 20 neurons; right, 9 neurons) versus stimulation rate; dotted lines, the type-defined populations used in this study. **c**, Active neurons when each of the 24 right sugar GRNs is omitted in turn from 100-Hz stimulation; omitting one LB3b neuron (green) abolishes the runaway. Dotted line, runaway criterion. **d**, Synapses from each left and right LB3b neuron onto antennal-lobe local neurons; bars, medians; green, the neuron identified in **c**.